Advanced Robot Agents: Foundation Models, RAX, ASPIRE, and Continual Improvement

Contents

Course home ·中文 Markdown ·English PDF

Foundation of Embodied AI ·Revised course notes

From general-purpose foundation models to online exploration, verifiable execution, and reusable robot skills

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> **Central questions:** How can Claude, Codex, and Gemini 4 Argon participate in robotics without being real-time robot controllers? Where does the reasoning in RAX and ASPIRE come from? How do foundation models, tool calls, physical feedback, and reusable programs combine into a useful, trustworthy robot agent?

Learning outcomes: Distinguish foundation models, coding agents, agent harnesses, robot agents, VLAs, and low-level controllers; understand how a general model processes observations, selects tools, writes programs, and incorporates execution feedback; explain RAX's Explore–Compile–Evaluate cycle, ASPIRE's trace-guided refinement, and the publicly demonstrated robotic applications of Gemini 4 Argon; and understand the responsibility and safety boundaries of a unified system. **No code listings or laboratory exercises are included.**

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1. Robot Agents: Definition and Responsibility Boundaries

1.1 From action prediction to task-level closed-loop reasoning

Suppose the robot is instructed to "locate the screwdriver, pick it up, move it to the target area, and rotate it by 90 degrees." A VLA or action policy mainly decides **what action to execute next**. A robot agent makes decisions at another level: **what should happen first, which tool should be called, how to check whether it worked, and what to change after a failure**. These are complementary components, not mutually exclusive alternatives.

$$a_t = \pi_\theta(o_t, g), \qquad \mathcal{P}_{k+1} = \mathrm{Agent}_M(g, \mathcal{H}_k, \mathcal{T}, \mathcal{E}_k).$$

Here o_t denotes sensory observations, g the task goal, and a_t an action-policy output. M is the foundation model, $\mathcal{H}_k$ the interaction history, $\mathcal{T}$ the available tools, $\mathcal{E}_k$ physical feedback, and $\mathcal{P}_{k+1}$ an updated plan or program. These expressions are **conceptual teaching abstractions**; they do not assert that RAX, ASPIRE, and Argon share an optimization objective.

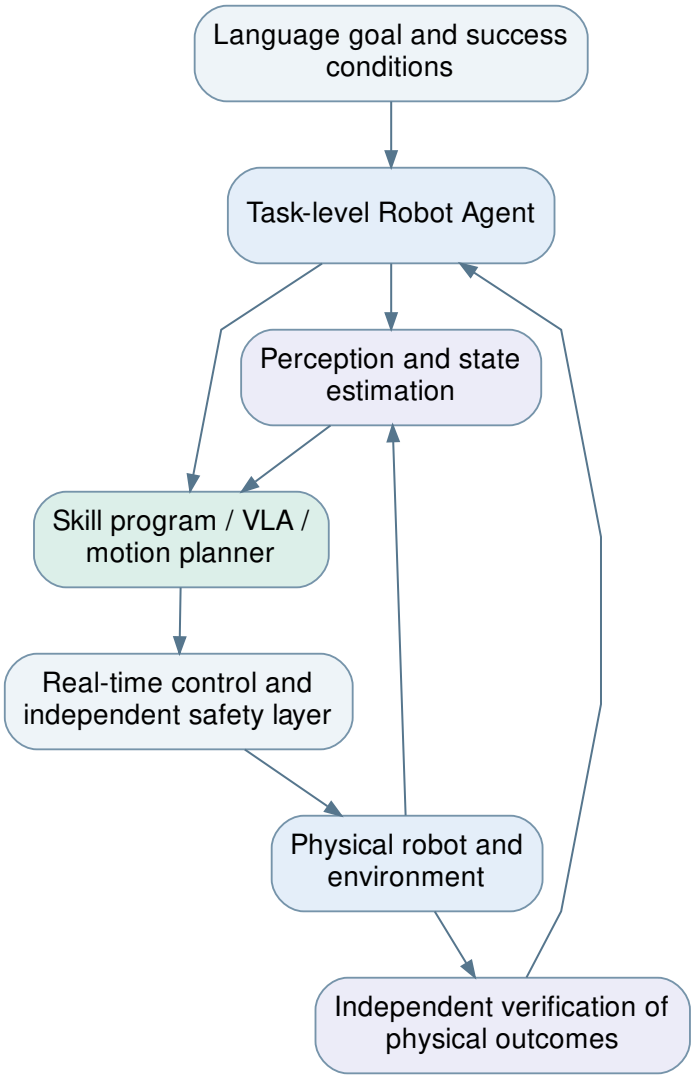


Figure 1. A robot agent manages task-level decisions, while physical feedback and safety must remain closed-loop. The arrows show functional relationships; they do not imply that each component is a separately trained large model.

Layer	Main responsibility	Typical timing and output
Task-level agent	Goal decomposition, tool choice, skill retrieval and generation, failure diagnosis, replanning	Event-driven; plans, programs, tool calls
Perception and skill execution	Object localization, state estimation, conditional branching, grasp/manipulation policy	Observation or skill steps; poses, action chunks
Motion control and safety	Joint/end-effector tracking, contact stabilization, constraints, emergency stop	High-rate feedback; position, velocity, or torque commands

Key principle: A foundation model may reason slowly; a robot cannot wait for it at every servo update. The safety and low-level feedback loops must not depend on a cloud model's latency or correctness.

1.2 A robot agent is neither a VLA nor a classical planner

- **VLA / action policy**: Maps observations and language to motions or action chunks and is well suited to continuous manipulation. Long-term memory, explicit tool use, and failure attribution depend on additional system design rather than being universal VLA properties.
- **Classical task and motion planning**: Reasons explicitly about geometry, collision, reachability, and temporal constraints. An agent can complement it when the goal is semantically novel or the failure is not covered by the modeled problem.
- **Coding robot agent**: Composes perception, simulation, planning, and robot APIs into programs and repairs those programs from physical feedback. Language reasoning alone supplies neither contact stability nor safety guarantees.

1.3 What does task success mean?

A motion API returning `FINISHED` only reports command completion. A closed gripper does not establish a grasp, and a wrist rotation does not prove that a screwdriver rotated. A stronger design independently checks physical postconditions:

$$V(o_{0:T}) \in \{\text{SUCCESS}, \text{FAILURE}, \text{UNKNOWN}\}.$$

`UNKNOWN` represents insufficient evidence: occlusion, pose-tracking drift, or an unobservable contact state. The proper next step may be active re-observation or a safe pause, not silently treating "no reported failure" as a success.

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2. How Claude, Codex, and Gemini 4 Argon Participate

2.1 Distinguish the model from the agent system

Claude denotes a foundation-model family, and Gemini 4 Argon is a general-purpose multimodal reasoning model. Claude Code and Codex, by contrast, are agentic coding products and harnesses through which models can inspect files, execute commands, use tools, and handle feedback. Calling all three "foundation models" is imprecise. The name Codex can also occur in model-related contexts; throughout this chapter, **Codex means the coding-agent product and harness**. [5–7]

Name	Technical layer	Typical role in robotics	Common misconception
Claude	General-purpose foundation model	Reasoning over text/images, planning, program generation, failure interpretation	A model API call does not automatically control a robot
Claude Code	Coding agent and harness	Workspace access, MCP/command and robot-tool calls, iterative edits and verification	It is not the Claude model weights themselves
Codex	Model-powered coding agent and harness	Tool use, program generation and repair, sandboxing and approval	The product is not interchangeable with one specific base model
Gemini 4 Argon	General-purpose multimodal reasoning model	Physical/video reasoning, robotic planning, code generation, Real2Sim	It is not inherently a fixed-rate joint-action VLA
Google Antigravity / custom harness	Agent execution environment	Connects Argon to context, tools, and execution feedback	Argon can be deployed through different environments

Name	Technical layer	Typical role in robotics	Common misconception
RAX / ASPIRE	Robot exploration and learning workflows	Organize exploration, verification, debugging, and memory	Neither is a newly trained universal foundation model

2.2 Six activities a foundation model can support

Within a robotic task, the model can help **(1) interpret the language goal, (2) reason about images, video, and structured state, (3) select a tool call, (4) synthesize a conditional skill program, (5) diagnose a failed trace, and (6) summarize reusable experience**. Each activity still requires an agent runtime and robot interfaces. A model's ability to *write* a program is not equivalent to a system's permission or ability to *safely execute* that program on hardware.

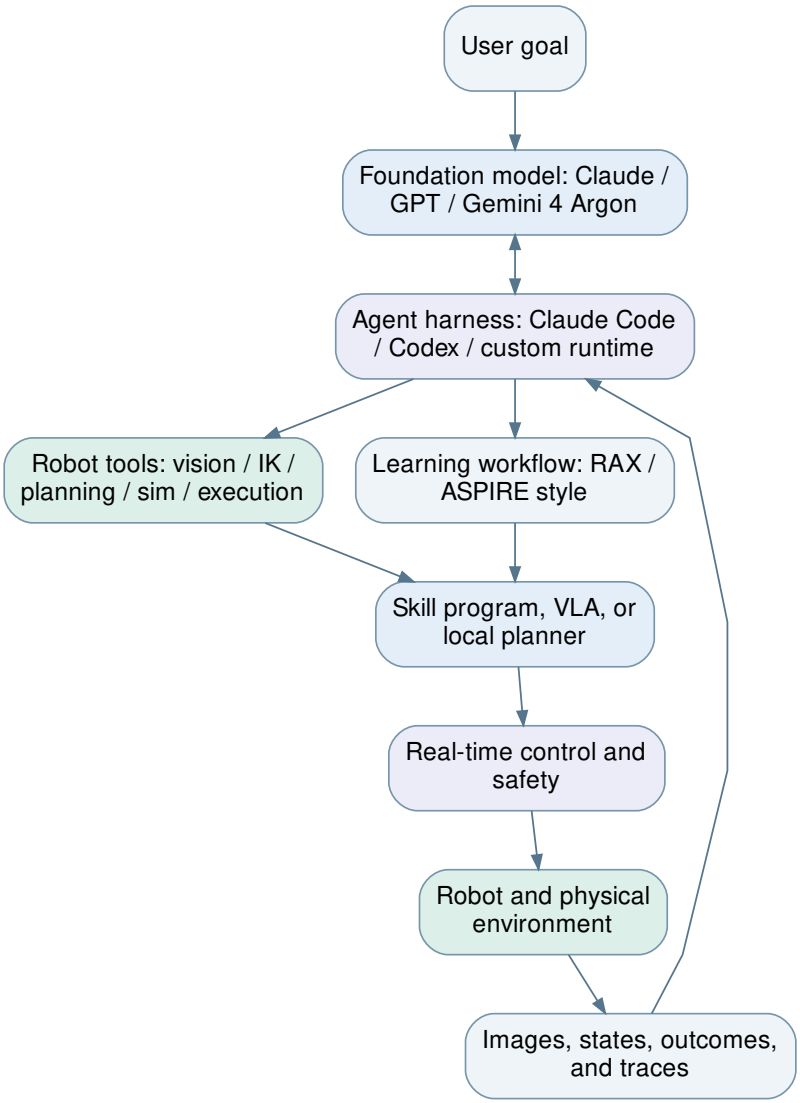


Figure 2. The foundation model supplies reasoning, the harness implements the agent loop, robot tools expose action

channels, and RAX/ASPIRE specify how experience is explored, tested, and accumulated. This is an **illustrative reference architecture**, not a claim that all three systems have been integrated in a published implementation.

2.3 Why the harness matters as much as the model

A harness is more than a prompt template. It manages **context, workspace state, tools and permissions, the interaction loop, execution traces, timeouts, failures, approvals, and termination conditions**. OpenAI documents Codex as an execution system between models and tools; Anthropic documents Claude Code's MCP, Skills, Hooks, and Agent SDK extension points. [5,6]

Calling the same base model once for an answer and embedding it in an iterative loop of **observe → invoke tool → inspect output → choose next action** are substantially different capabilities. Still, harnesses do not eliminate model errors: robot state, hard constraints, and success verification should come from externally checkable mechanisms.

2.4 Three practical technology paths

Claude path. A Claude model accesses restricted robot services through Claude Code or a custom agent runtime. RAX-style exploration can be compiled into a stand-alone procedure, while ASPIRE-style traces support targeted repairs and reusable lessons. [2–5]

Codex path. Codex already offers a coding-agent loop for reading workspaces, calling tools, generating programs, and iterating on feedback. The RAX implementation explicitly supports Codex CLI or Claude Code, while ASPIRE distributes agent skills compatible with Codex. [2,4,6]

Argon path. Gemini 4 Argon can reason about robots and simulation through external tools and runtimes. Google has demonstrated robotics workflows involving control, Real2Sim, video analysis, and program improvement. Those demonstrations **do not establish that Argon is integrated into the official RAX/ASPIRE implementations or can replace servo controllers**. [7,8]

General lesson: Selecting the model is only one design decision. What matters just as much is **which robot capabilities are exposed, how feedback is represented, how dangerous calls are prevented, and where verified experience is stored**.

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3. A General Foundation-Model-Driven Robot-Agent Workflow

3.1 From a goal to a bounded execution

A useful lifecycle is:

1. **Specify:** Convert natural-language goals into observable postconditions; define the workspace, budget, and safe-stop rules.
2. **Observe:** Read RGB/RGB-D, joint state, known geometry, and planner feedback; preserve confidence and uncertainty.
3. **Reason & Plan:** Use a foundation model through the harness to choose an existing skill, tool call, or procedure to explore.
4. **Act:** Call approved robot services, IK/planning, or action policies; let the local controller track continuous motion.
5. **Verify:** Check the actual physical goal rather than the completion flag of a command.
6. **Repair & Remember:** Inspect failure traces, apply bounded patches, independently retest, and retain lessons backed by evidence.

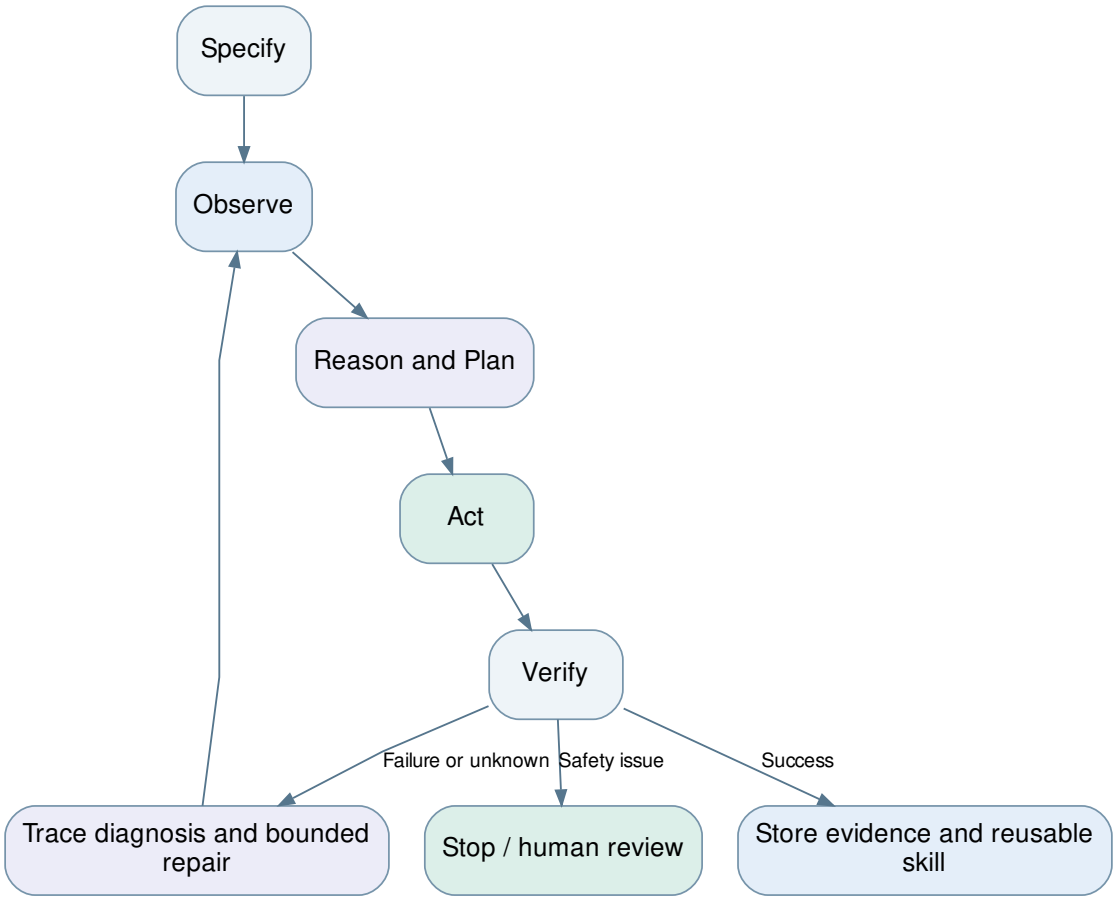


Figure 3. The observe–decide–act–verify loop may repeat, but retries need explicit budgets, an independently enforced safety boundary, and an intelligible stopping condition.

3.2 Tool calling connects reasoning to physics; it is not a controller

Tool group	What an agent may request or receive	Required interface behavior
Sensing and state estimation	Images, detections, object poses, confidence	Timestamps, reference frames, uncertainty and failure flags
Geometry and planning	IK solutions, collision checking, candidate approaches	Reachability, kinematic constraints, collision enforcement
Skill or policy execution	Grasp, transport, rotation, VLA/RL action chunks	Preconditions, bounds, timeout and local recovery
Simulation and evaluation	Replay, physical simulation, comparisons, task scores	Explicit sim-to-real limits; no conflation of sim and hardware success
Logging and memory	Multimodal traces, program versions, applicability	Provenance, rollback, auditable records

3.3 Four loops should not be collapsed into one "super-model"

The **control loop** handles rapid feedback and stability; the **skill loop** handles observation updates, action sequencing, and limited local recovery; the **task-agent loop** handles slower replanning, coding, and diagnosis; and the **long-term-learning loop** validates, versions, reuses, and retires skills. These are functional loops, not requirements for four distinct foundation models. For a previously validated task, it may be preferable to execute the compiled skill without invoking the large model again.

3.4 What is learned? Program memory is not weight fine-tuning

RAX and ASPIRE primarily improve the system by accumulating **programs, failure explanations, skill-library entries, and execution evidence**; they do not require online gradient updates to foundation-model weights. A different route uses simulation-generated demonstrations to train or post-train a VLA. In that case, the **action model's weights** change in a separate training stage. Robot interaction, self-improvement, online RL, and foundation-model fine-tuning are therefore **not synonyms**.

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4. RAX: From Exploration to Stand-Alone Programs

RAX (Robot Agent eXploration) lets a general-purpose coding agent with **frozen model weights** (the reference harness supports **Codex CLI or Claude Code**) explore through perception and control tools, keep what it learns in a workspace, and compile effective behavior into modular robot code that runs without continuous coding-agent intervention. Its organizing cycle is **Explore → Compile → Evaluate**. RAX does not retrain the language model for every newly encountered task. [1,2]

4.1 Explore: learn conditional decisions, not motor physics again

The agent can investigate which camera viewpoint to use, how to approach an object, whether a grasp direction is reachable, and when to confirm that the object is still held. Each attempt should preserve **the observation, invoked tool and parameters, error/output, physical outcome, and reason for a change**. A log that says only "grasp failed" transfers little. A log that says "the approach fell inside the table collision margin; a lateral approach then planned successfully" contains a reusable causal clue.

4.2 Compile: a successful trajectory is not yet a reusable program

One successful motion replay	RAX-style compiled procedure
Reuses object coordinates from exploration	Re-observes and recomputes goals in the current scene
Executes a fixed action list	Branches on IK, contact, and visual feedback
Stops or asks the agent to rescue every failure	Includes bounded recovery and explicit abort conditions
Establishes feasibility for one setup	Declares preconditions and is validated on changed layouts

In RAX, **compilation means synthesizing independent, conditional robot logic from exploration experience**, not translating a language prompt into processor instructions. The reference project uses `controller.py:main(ctx)` as an example program entry point; its name is cited to clarify the software boundary, not to introduce a coding assignment. [2]

4.3 Evaluate: validation separate from the exploratory agent

An agent that remains active during debugging might silently compensate for missing code. RAX therefore freezes the generated program and evaluates it independently on held-out simulated arrangements or reset hardware runs. Distinguish **one success**, **repeatability under identical conditions**, and **unattended success after initial-state variation**: they provide increasingly strong but still bounded evidence. The RAX project's air-fryer example reports roughly **72 minutes of exploration** and **18 seconds of compiled execution**. These are task-specific figures, not universal real-time-performance guarantees. [1,2]

The official skill library additionally records preconditions, code, known failure modes, independent run records, and camera evidence. A working script should become a *verified skill* only under explicit review and repeat-test rules. [2,3]

Chapter takeaway: RAX shows how methods discovered online can become **reusable, observation-dependent programs that no longer require step-by-step language-model control**.

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5. ASPIRE: Trace-Guided Repair and Accumulating Skills

ASPIRE also follows **code-as-policy**: a coding agent writes a robot program and revises it from physical or simulated execution feedback. Its distinctive elements are **multimodal execution traces, iterative debugging, growing skill libraries, Coordinator–Actor organization, and evolutionary program search**. The original studies used Claude Opus 4.6, while the maintained implementation includes agent skills for environments such as Claude Code and Codex. The workflow should not be equated with a single underlying model. [4,9]

5.1 Traces: from an error label to a causal bottleneck

Trace evidence	Question it answers	Likely diagnostic direction
Tool call and parameters	Was the pose, grasp direction, or constraint wrong?	Check parameters and coordinate transforms
Planner return and error	Did IK, reachability, or collision checking fail?	Choose another approach or replan
Keyframes and estimated poses	Did the object physically move, or was the grasp empty?	Re-observe and adjust contact timing
Contact/proprioceptive timeline	Was there slip, or only estimator drift?	Prioritize the best-supported failure cause
Verification outcome and confidence	Did the physical goal actually occur?	Seek another viewpoint or return `UNKNOWN`

A useful agent selectively inspects **evidence around the failing primitive** instead of filling its context window with every raw video frame. When the planner reports `PLANNING_ERROR`, inspecting the target, collision margin, and route is more relevant than blindly increasing grip force.

5.2 Iterative debugging: each patch must be testable

One repair iteration follows **select trace → formulate a local hypothesis → patch the program → rerun → compare evidence**. The ASPIRE paper's radio-retrieval example illustrates the logic: perception found the object, but navigation candidates conflicted with collision margins near a table. Changing the approach direction enabled the task to proceed. The generalizable lesson is not to add one fixed displacement everywhere, but to **search for a safe alternative viewpoint or approach when the intended route is obstructed**. [9]

5.3 Skill library, Coordinator–Actor, and evolutionary search

- **Skill library**: Retain verified programs, reusable repair patterns, applicability conditions, and supporting evidence. One accidental success is not a universal rule.
- **Coordinator–Actor**: Organize multiple coding agents across tasks; check whether candidate solutions respect interfaces and actually work before sharing them.
- **Evolutionary search**: Try alternative candidate programs or decision sequences, score their executions, and use traces to choose further improvements. The evolved objects are **programs and task logic**, not necessarily neural-network weights.

Chapter takeaway: ASPIRE turns failures into diagnosable evidence, makes repairs independently testable, and helps transferable skills accumulate across tasks.

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6. Gemini 4 Argon: Control, Real2Sim, and Self-Improvement

On **October 13, 2026**, Google published a collection of robotic demonstrations using Gemini 4 Argon. Argon is a **general-purpose reasoning foundation model** that can process visual and video inputs and use tools. Once integrated with robot APIs, simulation tools, and repeated observations, the resulting system can act as a **robot agent**. The model itself is not a servo-rate action policy. [7,8]

6.1 Four categories demonstrated by Google

Application	Published demonstration	What the model contributes
Robot control	Spot manipulation with MuJoCo practice; uprighting a mug and inserting a napkin	Decomposed tool actions, visual checking, recovery from feedback
Real2Sim	Building MuJoCo scenes from robot video, egocentric human video, or sketches	Scene/asset synthesis and iterative visual-contact consistency checks
Physical understanding	Video hindsight labeling, contact-level events, task-progress estimation	Structured process labels and evidence useful for evaluation
Research and improvement	Simulation-generated action datasets; SO-101 hardware improvement	Better programs, environments, and data-generation pipelines

6.2 How to interpret the case studies

Case A: Visual closed-loop manipulation. In the mug-and-napkin task, Argon uses a calibrated wrist-camera representation and a set of existing action primitives. It selects the next operation and checks feedback after tool calls. This is **model reasoning + harness + robot tools**, not direct neural control of motor current. [8]

Case B: Real2Sim. Google shows robot-teleoperation footage being used to construct a MuJoCo scene, tune object geometry, mass, friction, and other contact-related properties, and compare rendered views and force/tactile feedback. This is a **scene reconstruction and physics-consistency workflow**. Visual similarity does not, by itself, prove exact contact identification or general sim-to-real transfer. [8]

Case C: Data generation and post-training. Argon can help build simulation tasks and generate trajectories subsequently used to improve a VLA. In that pipeline, **Argon constructs simulation/data, while any VLA parameter update is a separate learning process**. It would be incorrect to conclude that Argon's own weights were updated by online reinforcement learning. [8]

Case D: Hardware self-improvement. In an SO-101 visual pick-and-place task, Google reports three execution iterations taking **58.9, 22.9, and 17.4 seconds**, respectively. This documents one showcased agentic optimization sequence, not a general statistical guarantee for robot success or reliability. [8]

6.3 Complementarity with RAX and ASPIRE

System	What it is	Primary research emphasis	What it does not establish
Gemini 4 Argon	General multimodal foundation model	Physical reasoning, code generation, video understanding	Native high-rate servo control
RAX	Online exploration and program-compilation framework	Stand-alone execution after skill discovery	Requirement for a newly trained foundation model
ASPIRE	Trace-based debugging and continual skill-refinement framework	Failure diagnosis and transferable repair	VLA parameter post-training by definition

Conceptually, Argon could serve as a reasoning model in a RAX-like or ASPIRE-like architecture, but **a plausible combination is not an experimental result published by those projects**. Useful comparisons must separate the model's capability, the harness, the exposed robot API, and the independently verified physical outcome.

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7. A Unified System and a Dexterous Screwdriver Example

7.1 One reusable workflow without overstating the papers

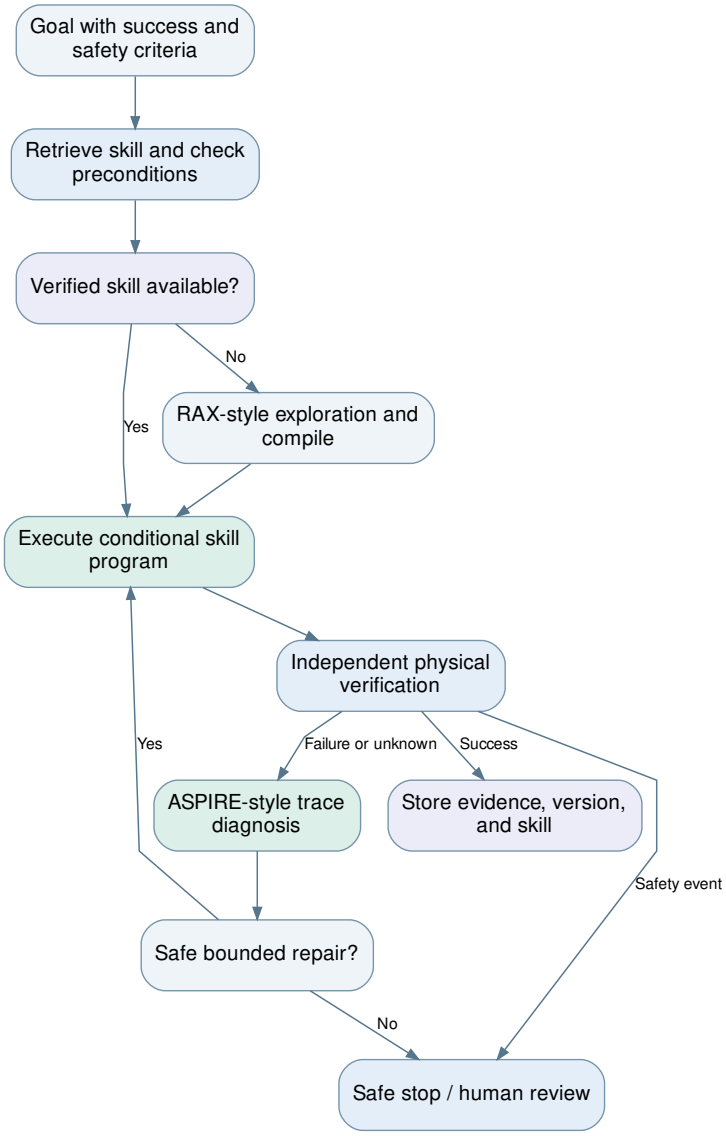


Figure 4. RAX-style exploration fills skill gaps; ASPIRE-style debugging and memory address recurrent failures; Claude, Codex, or Argon-related model/harness components can provide higher-level reasoning. This is a **teaching reference architecture**, not a system whose joint performance has already been established by the cited papers.

7.2 The minimum contract for a reusable skill

1. **Preconditions**: Robot body, dexterous hand, sensing/calibration assumptions, object class, and permitted workspace.
2. **Execution**: Parameter interface, approved tool calls, duration/budget, and delegated low-level control.
3. **Verification**: Object-state postconditions, tolerance, and `SUCCESS / FAILURE / UNKNOWN` criteria.
4. **Recovery**: Recoverable failures, retry limits, and safe exit for unrecoverable cases.
5. **Evidence & Version**: Independent run records, failure-distribution limits, version history, and rollback path.

This is an **engineering recommendation**, not a universal data structure prescribed by one paper. A script should not be promoted to a deployable skill without constrained independent tests and the required approvals.

7.3 Worked example: grasp, transport, and rotate a screwdriver

Illustrative teaching scenario, not an experiment reported in the cited papers. A screwdriver starts at an arbitrary permitted location on a table. A robot observes it through head and wrist cameras, uses a dexterous hand to grasp it, transports it to a fixed region, and rotates it through a desired angle. If it slips, dangerous motion must stop before the agent decides whether re-localization and recovery are safe.

Stage	Physical evidence required	Common false success and response
Locate	Correct screwdriver identity, usable pose, and confidence	Misidentified tool or wrong camera frame: reconfirm
Grasp	Object leaves the table and its hand-relative pose is plausible	Fingers close on empty space: compare vision and proprioception
Transport	Object reaches the goal zone and remains held	End-effector arrives without object: stop and re-localize
Rotate	**Object itself**, not just wrist, attains the target orientation	Independent object-pose tracking is necessary
Recover	New location is authorized and the goal is re-established	Unobservable or unreachable: return `UNKNOWN` or stop

What does the task-level agent do? It specifies observable success, searches for existing skills, explores missing grasp/approach decisions using RAX-like tools, compiles conditional procedures, diagnoses failures from ASPIRE-like traces, and allows bounded repairs. Claude, Codex, or Argon may provide the reasoning and programming capabilities, **but this example does not assert that any of these particular models has been experimentally validated on this hand/arm platform**.

What does the lower stack do? Perception estimates hand/object relative poses; IK and motion planning enforce reachability and collision conditions; VLA/RL skills or local controllers perform grasping and rotation; independent hardware safety enforces limits and emergency stops. With only head/wrist cameras and no tactile sensor, occlusion and unobservable contact must be represented explicitly in verification outcomes.

7.4 Which component should solve which problem?

Practical issue	First component to consider	Main evaluation criterion
Complex semantic goal, video interpretation, program synthesis	**Claude / Gemini 4 Argon or another foundation model**	Suitability of physical reasoning and program generation
Persistent tool use, workspace handling, approval and logging	**Claude Code / Codex / custom harness**	Permissions, state management, auditability, failure recovery
Novel task without an existing executable skill	**RAX-style exploration and compilation**	Conditional procedure with independent validation

Practical issue	First component to consider	Main evaluation criterion
Recurrent failures whose causes are unclear	**ASPIRE-style traces and skill library**	Evidence-supported diagnosis, localized repair, transferable knowledge
Continuous contact or high-rate trajectory tracking	**VLA / RL / MPC / real-time controller**	Latency, stability, constraints, physical safety

Final takeaway

A robot agent is not simply a larger VLA or a model that can write robot code. It is a task-level closed-loop system combining **foundation-model reasoning**, **an agent harness**, **robot tools**, **execution feedback**, **verifiable skill memory**, **and independently enforced safety**. Claude and Gemini 4 Argon may supply general reasoning, Claude Code and Codex provide agent execution environments, RAX emphasizes compiling exploration into independently runnable programs, and ASPIRE emphasizes trace-based repair and skill accumulation. Whether a real robot works reliably ultimately depends on **observability**, **tool quality**, **physical verification**, **and low-level control**, not on the model's name alone.

Papers and Official Resources (Clickable)

1. **RAX paper**: Huang et al., *Scaling Online Exploration for Robot Agents* (2026). Paper PDF ·Project.
2. **RAX reference harness**: robot-agent-exploration/rax, documenting Codex CLI/Claude Code, exploration, compilation, and independent evaluation.
3. **RAX skill library**: robot-agent-exploration/skills-lib, with procedures, preconditions, and execution evidence.
4. **ASPIRE reference implementation**: NVlabs/ASPIRE, including iterative debugging, agent skills, and evolutionary search.
5. **Claude Code documentation**: Anthropic, Claude Code Overview, covering the agent runtime, MCP, and tool integrations.
6. **Codex harness**: OpenAI, *Codex as a platform: build on the open agent harness* (2026).
7. **Gemini 4 Argon model**: Google AI for Developers.
8. **Gemini 4 Argon for Robotics**: Google Developers Blog (Oct. 13, 2026); source for the Argon robotic demonstrations discussed here.
9. **ASPIRE paper**: Lu et al., *ASPIRE: Agentic /Skills Discovery for Robotics* (2026). arXiv ·Project page.

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